

Dyslipidemia

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Outlines

- Introduction
- Etiology
- Risk factors
- Pathogenesis
- Clinical presentation
- Diagnosis
- Treatment

Introduction

- **Dyslipidemia** can be defined as elevated total cholesterol, LDL-C, or triglycerides level, low HDL-C, or some combination of these abnormalities.
- Total cholesterol and LDL-C increase throughout life in men and women
- Abnormalities of plasma lipids can result in a predisposition to **coronary, cerebrovascular, and peripheral vascular arterial disease.**

Introduction

- **Cholesterol, triglycerides, and phospholipids** are transported in the bloodstream as complexes of lipid and proteins known as **lipoproteins**.
- Plasma lipoproteins - ->spherical particles with
 - surfaces that consist:** phospholipid, free cholesterol, and protein
 - cores that consist:** triglyceride and cholesterol ester

Introduction

- Based on their density and composition: plasma lipoproteins are classified as:
 - Chylomicrons
 - Very-low-density lipoproteins (VLDL)
 - Low-density lipoproteins (LDL)
 - High-density lipoproteins (HDL)
- Density of lipoproteins
 - Determined by their relative content of protein and lipid

Introduction

- HDL transports cholesterol from lipid-laden foam cells to the liver (**happy/good Cholesterol**)
- LDL transports fat from liver to body (**bad cholesterol**)
- **VLDL**, the major lipoprotein associated with triglycerides, is enriched with cholesterol esters
 - VLDL is carried in the circulation as triglyceride (TG)

- **Chylomicrons** are large triglyceride-rich particles that contain apolipoproteins (protein constituent of lipoproteins) B-48, B-100, and E.
 - They are formed from dietary fat
- **VLDL** synthesis is regulated in part by **diet** and hormones
- VLDL is secreted from the liver and serially converted to IDL and finally to LDL
- VLDL and LDL are calculated values:
 - $\text{VLDL} = \text{TG}/5$
 - $\text{LDL} = \text{Total cholesterol} - (\text{VLDL} + \text{HDL})$.

Function of Fats in the body

- Cholesterol & triglycerides are essential substrates for cell membrane formation
- Hormone synthesis
- They provide a source of free fatty acids
- **The rate-limiting enzyme in the cholesterol synthesis pathway is 3-hydroxy-3-methylglutaryl coenzyme A reductase (HMG-CoA reductase)**

Etiology

- Generally, Genetic and environmental factors
- Primary or genetic/familial lipoprotein disorders classified in to six
 - **I** (chylomicrons), **IIa** (LDL),
 - **IIb** (LDL + VLDL), **III** (IDL),
 - **IV** (VLDL), and **V** (VLDL + chylomicrons)
- Secondary forms of hyperlipidemia also exist.
- “The 4D classification” of secondary causes of dyslipidemia include: **diet, drugs, disorder, and diseases.**
- Regarding diet
 - excessive alcohol use, weight gain, excessive carbohydrate intake, and high saturated fat intake.

Etiology

- **Certain drugs**
 - atypical antipsychotics, diuretics, beta blockers, glucocorticoids, oral estrogen and progestin, tacrolimus, and cyclosporine.
- **Certain metabolism disorders**
 - **Nephrotic syndrome, renal failure, biliary obstruction, hypothyroidism, and pregnancy**
- **Comorbid conditions or diseases**
 - obesity, **uncontrolled diabetes, liver disease, chronic kidney disease**
- Although the lipid disorders are classified into primary and secondary based on etiologies, most dyslipidemias are a result of a combination of both.

Risk Factors

The Helsinki Heart Study

- Hypertriglyceridemia and low HDL are associated with:
 - ✓ Obesity (body mass index [BMI] >30 kg/m²)
 - ✓ Smoking
 - ✓ Sedentary lifestyle
 - ✓ Blood pressure 140/90 mm Hg???? (HTN)
 - ✓ Blood glucose HgA1C >4.4 ??? (DM)
 - ✓ Alcohol intake

Pathogenesis of Atherosclerotic Cardiovascular Disease

- **The “Response-to-injury” hypothesis states that risk factors such as:**
 - **Elevated Oxidized LDL**
 - Mechanical injury to the endothelium
 - Excessive homocysteine,
 - Immunologic attacks, or
 - Infection-induced changes in endothelial and intimal function
- **Lead to endothelial dysfunction and a series of cellular interactions that culminate in atherosclerosis.**

- Atherosclerotic lesions are thought to arise from transport and retention of plasma LDL-C through the endothelial cell layer into the extracellular matrix of the sub-endothelial space
- Lipid abnormalities increase the risk for **CHD and cerebro-vascular morbidity and mortality.**
- The eventual outcomes of this atherogenic cascade are clinical events such as:
 - angina, MI, arrhythmias, stroke, peripheral arterial disease, abdominal aortic aneurysm, and sudden death.

Clinical Presentation

- Mostly asymptomatic for many years

Symptoms:

- None to **severe chest pain, palpitations, sweating, anxiety, shortness of breath, loss of consciousness or difficulty with speech or movement, abdominal pain, sudden death.**

Signs:

- None to severe **abdominal pain, pancreatitis, eruptive xanthomas, peripheral polyneuropathy, high BP, BMI >30 kg/m² or waist circumference >40 inches in men >35 inches in women**

Laboratory Tests

- Elevations in
 - Total cholesterol
 - LDL
 - Triglycerides
 - Apolipoprotein B
 - C-reactive protein
- Low HDL (< 40mg/dl)

Patient Evaluation

- **A fasting lipoprotein** profile including total cholesterol, LDL-C, HDL-C, and triglycerides should be measured in all adults 20 yrs and older at least once every 5 yrs.
- If total cholesterol is > 200 mg/dL or HDL-C is < 40 mg/dL in non-fasting state , a follow up fasting lipoprotein profile is recommended.

TABLE 9-1**Classification of Total, LDL, and HDL Cholesterol and Triglycerides**

Total cholesterol	
<200 mg/dL	Desirable
200–239 mg/dL	Borderline high
≥240 mg/dL	High
LDL cholesterol	
<100 mg/dL	Optimal
100–129 mg/dL	Near or above optimal
130–159 mg/dL	Borderline high
160–189 mg/dL	High
≥190 mg/dL	Very high
HDL cholesterol	
<40 mg/dL	Low
≥60 mg/dL	High
Triglycerides	
<150 mg/dL	Normal
150–199 mg/dL	Borderline high
200–499 mg/dL	High
≥500 mg/dL	Very high

- A complete hx and physical examination should assess:
 - (a) CV risk factors or definite CVD
 - (b) Family hx of premature CVD or lipid disorders
 - (c) secondary causes of lipid abnormalities (drugs)
 - (d) Xanthomas or abdominal pain; hx of pancreatitis, renal or hepatic disease, peripheral vascular disease, abdominal aortic aneurysm, or cerebral vascular disease

Major CV Risk Factors

- Age
 - Men 45 years
 - Female 55 years
- Premature menopause without estrogen replacement therapy
- Family history of premature CHD (before 55 years of age for male and 65 years of age for female first degree relatives)
- Cigarette smoking
- Hypertension ($> 130/80$ mm Hg or on antihypertensive medications)
- Low HDL (< 40 mg/dl)
- DM- CHD equivalent

Risk stratification

- Four categories of risk that modify the goals and modalities of LDL lowering therapy
 - Highest Risk** : CHD or CHD risk equivalents,
 - Risk for major coronary events equal to or greater than established CHD
 - >20% per 10 years (2% per year)
 - Moderately high risk**:
 - patients with multiple (2+) risk factors
 - 10-year risk for CHD is 10% to 20%

Risk stratification

– Moderate risk

- ≥ 2 risk factors and
- a 10-year risk of $\leq 10\%$

– Lowest risk

- Persons with zero to one risk factor
- Risk is estimated from Framingham risk scores based on patient's age, LDL-C or total cholesterol level, blood pressure, the presence of diabetes, and smoking status

Treatment

Goals of Therapy

- To reduce the risk of first events or recurrent events such as MI, CHD, heart failure, ischemic stroke, and
- peripheral arterial disease, such as carotid stenosis and abdominal aortic aneurysm

Goals of therapy

Risk Category	LDL Goal (mg/dL [mmol/L])	LDL Level at which to Initiate TLC (mg/dL [mmol/L])	LDL Level at which to Consider Drug Therapy (mg/dL [mmol/L])
High risk: CHD or CHD risk equivalents (10-year risk >20%)	<100 (<2.59) (optional goal: <70 [<1.81])	≥100 (≥2.59)	≥100 (≥2.59) (<100 mg/dL [<2.59]; consider drug options) ^b
Moderately high risk: 2+ risk factors (10-year risk >10–20%)	<130 (<3.36)	≥130 (≥3.36)	≥130 (≥3.36) (100–129 [2.59–3.35]: consider drug options)
Moderate risk: 2+ risk factors (10-year risk <10%)	<130 (<3.36)	≥130 (≥3.36)	≥160 (≥4.14)
Lower risk: 0–1 risk factor	<160 (<4.14)	≥160 (≥4.14)	≥190 (≥4.91) (160–189 [4.14–4.90]: LDL-lowering drug optional)

Non-Pharmacologic Therapy

Therapeutic Life Change (TLC)

- Implemented in all pts prior to considering drug therapy
- TLC may obviate the need for drug therapy, and allow for lower doses.
- TLC include:
 - **Dietary therapy: decrease intake of total fat, saturated fat, and cholesterol and Increase intake of soluble fiber, fish oil(omega3)**
 - **Increased physical activity**
 - **weight reduction**
 - **Stop smoking**

- Pts are given a 3-month trial (two visits 6 wks apart) of **TLC before advancing to drug therapy.**
 - Unless pts are at **very high risk**
 - Severe hypercholesterolemia
 - Known CHD
 - CHD risk equivalents
 - Multiple risk factors
 - Strong family hx

Fish Oil

- Fish oil supplementation provides an increased amount of the **omega-3 polyunsaturated fatty acids**, such as eicosapentaenoic acid and docosahexaenoic acid (**antioxidant effect**)
- Each 20 g/day ingestion of fish lowers CHD risk by 7%, and eating fish at least once weekly should reduce CHD mortality

Pharmacologic Therapy

- Reduction of LDL reduces CHD event rates in primary prevention and secondary intervention trials
- For every 1% reduction in LDL, there is a 1% reduction in CHD event rates
- A 1% elevation of HDL results in an approximately 2% reduction in CHD events

Drug Therapy

- Despite availability of many efficacious lipid-lowering drugs, none is effective for all lipoprotein disorders
- Classes of drugs
 - Statins
 - Bile acid resins (BARs)
 - Niacin
 - Fibrates

Pharmacologic Therapy

Lipoprotein Type	Drug of Choice	Combination Therapy
I	Not indicated	—
IIa	Statins Cholestyramine or colestipol Niacin	Niacin or BAR Statins or niacin Statins or BAR Ezetimibe Mipomersen, lomitapide ^a
IIb	Statins Fibrates Niacin	BAR or fibrates or niacin Statins or niacin or BAR ^b Statins or fibrates Ezetimibe
III	Fibrates Niacin	Statins or niacin Statins or fibrates Ezetimibe
IV	Fibrates Niacin	Niacin Fibrates
V	Fibrates Niacin	Niacin Fish oils

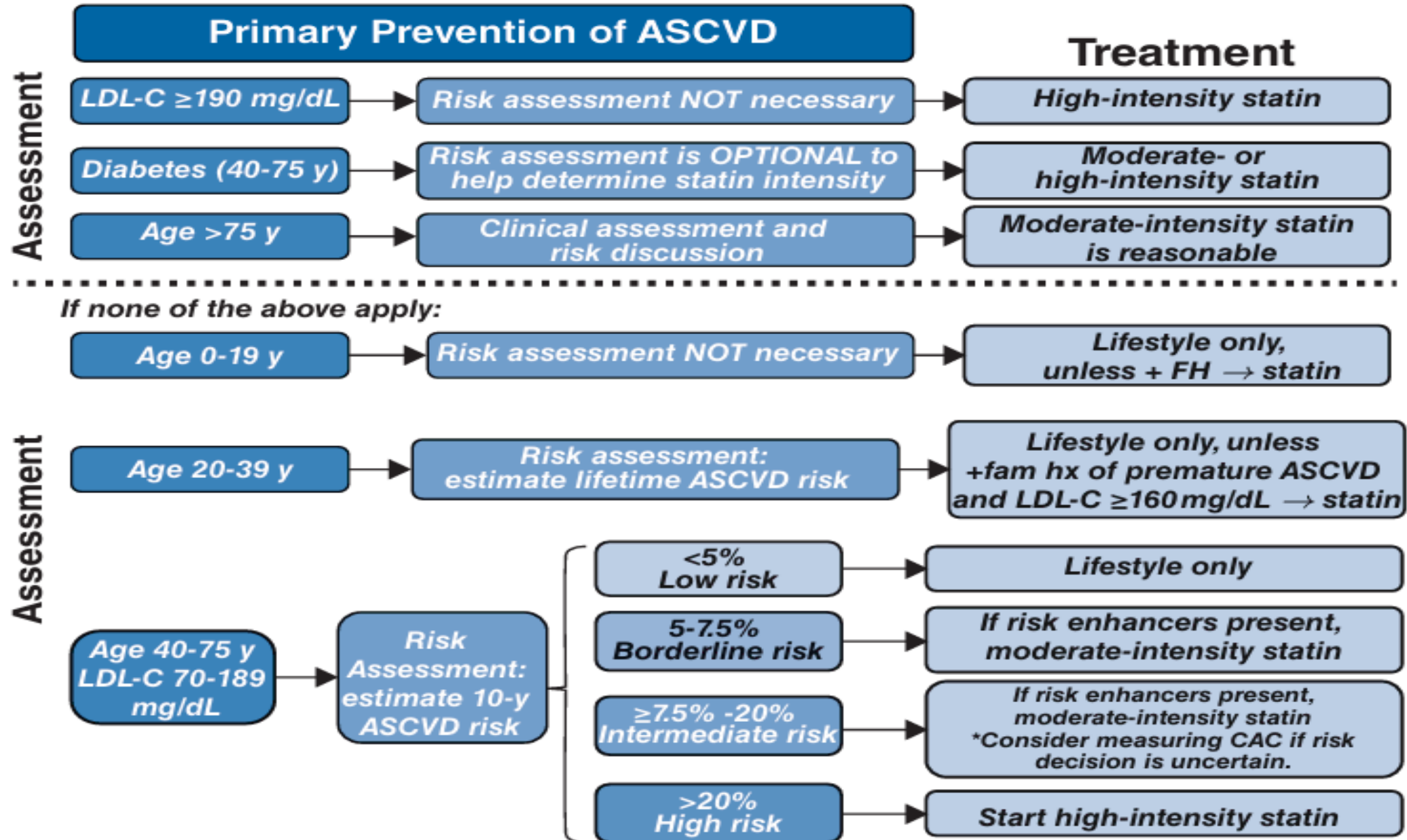
Statin Therapy

- **First choice** -most potent LDL-lowering agents
- They interrupt the conversion of **HMG-CoA to mevalonate, the rate-limiting step** in de novo cholesterol biosynthesis, by **inhibiting HMG-CoA reductase**
 - Lovastatin, pravastatin, simvastatin, fluvastatin, and atorvastatin
 - **Rosuvastatin** is the most potent statin currently on the market
- **TC & LDL-C** are reduced in a dose-related fashion when statins are added to dietary therapy

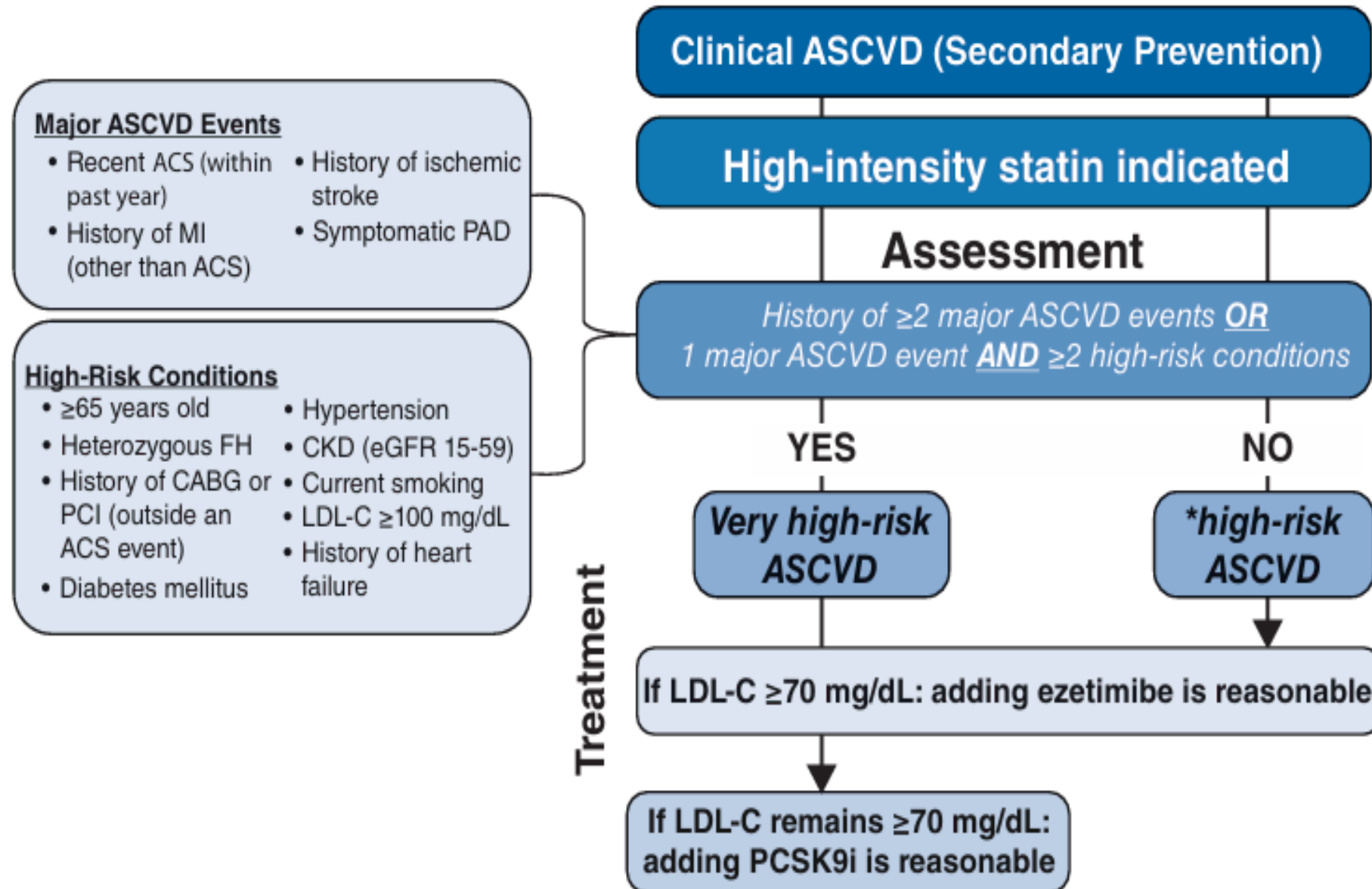
TABLE 31-7 Intensity of Statin Therapy by Drug and Dose

High-intensity Statin Therapy	Moderate-intensity Statin Therapy	Low-intensity Statin Therapy
Daily dose lowers LDL on average by $\geq 50\%$	Daily dose lowers LDL on average by 30% to <50%	Daily dose lowers LDL on average by <30%
Atorvastatin 40-80 mg	Atorvastatin 10-20 mg	Simvastatin 10 mg
Rosuvastatin 20-40 mg	Rosuvastatin 5-20 mg	Pravastatin 10-20 mg
	Simvastatin 20-40 mgⁱ	Lovastatin 20 mg
	Pravastatin 40-80 mg	Fluvastatin 20-40 mg
	Lovastatin 40 mg	Pitavastatin 1 mg
	Fluvastatin XL 80 mg	
	Fluvastatin 40 mg BID	
	Pitavastatin 2-4 mg	

Use of statins for primary ASCVD prevention



Use of statins for secondary ASCVD prevention.



Combinations

- **BARs plus lovastatin**
 - Further reduce enterohepatic circulation of bile acid
- **Statin plus Ezetimibe**
 - Ezetimibe inhibits cholesterol absorption across the gut border and adds 12% to 20% further reduction when combined with a statin or other drugs

Side Effects

- Elevation of LFT's (ALT) $> 3X$ the upper limit
- Serious muscle toxicity (**myopathy**) occurs in
 $<0.6\%$ of pts
- Lens opacities

Bile Acid Resins (BARs)

- **Include:** cholestyramine, colestevlam, colestipol
- The primary action of BAR is **binding bile** acids in the **intestinal lumen**, with concurrent **interruption of enterohepatic circulation** of bile acids and markedly increased excretion of acidic steroids in the feces
- Decrease **T-C & LDL-C** concentrations

BAR

- The increase in hepatic cholesterol **biosynthesis** may be increased in liver consequently, BARs may **aggravate hypertriglyceridemia** in pts with combined hyperlipidemia.
- **Side effects:** GI complaints of **constipation, bloating, epigastric fullness, nausea, and flatulence** are common
 - ✓ Managed by increasing fluid intake, increase bulk in diet, fiber foods & use stool softeners

- Other **potential adverse effects** include:
 - Impaired absorption of fat-soluble vit. A, D, E, & K
 - Hyponatremia and hyperchloremia
 - Gastrointestinal obstruction
- **Drug Interactions:** Reduced bioavailability of **acidic drugs** such as coumarin, digitoxin, nicotinic acid, thyroxine, acetaminophen, hydrocortisone, hydrochlorothiazide, loperamide, and possibly iron

BAR-adherence

- At least 40% of patients discontinue therapy within 1 year
- Adherence rates can be improved with pharmacist interventions
- Adverse effects can be managed by increasing fluid intake, increase bulk in diet, & use stool softeners

Niacin

- Reduces hepatic synthesis of VLDL, thus reduced synthesis of LDL
- Treatment of **primary hypercholesterolemia** alone or in combination with BARs
- Niacin also increases **HDL**
- Used primarily for treatment of mixed hyperlipidemia
- First-line agent or alternative for treatment of **hypertriglyceridemia** and **diabetic dyslipidemia**

NIACIN Side effects

- Cutaneous flushing and itching appear to be prostaglandin mediated
 - Reduced by **aspirin 325 mg** given shortly before niacin ingestion
 - Take dose with meals and slowly titrate the dose upward may minimize these effects

Niacin Side Effects

- **GI- intolerance and flushing** are common problems
- **Acanthosis nigricans**, a darkening of the skin in skinfold areas and an external marker of insulin resistance, may be seen with high doses of niacin
- Sustained-release products may minimize these complaints in some patients
- Elevated liver function tests, hyperuricemia, and hyperglycemia
- Preexisting **gout** and **diabetes** may be exacerbated

- Niacin is contraindicated in patients with active liver disease.
- Concomitant **alcohol** and **hot drinks** may magnify flushing and pruritus with niacin and should be avoided at the time of ingestion.

Fibrates

- **Fibric acid** (clofibrate , gemfibrozil, fenofibrate) monotherapy is **effective in reducing VLDL**,
- Gemfibrozil and fenofibrate are used much more commonly than clofibrate
- Gemfibrozil reduces **synthesis of VLDL** and, to a lesser extent, apolipoprotein B, with a concurrent increase in the rate of removal of triglyceride-rich lipoproteins from plasma

Fibrates

- Plasma HDL concentrations may rise 10% to 15% or more with fibrates
- **Side effects**
 - **Gastrointestinal complaints**, rash, dizziness and transient elevations in transaminase levels and alkaline phosphatase
 - Gemfibrozil and fenofibrate may enhance the formation of gallstones but the rate is low (0.5%–7%)

Fibrates

- Fibrates potentiate the effects of oral anticoagulants, so PT and INR should be monitored very closely
- A myositis syndrome of myalgia, weakness, stiffness, malaise, and elevations in creatinine phosphokinase and AST more common in pts with renal insufficiency

Ezetimibe

- **Ezetimibe** interferes with the absorption of cholesterol from the brush border of the intestine, a novel mechanism that makes it a good choice for adjunctive therapy.
- It is approved as both monotherapy and for use with a statin.
- The dose is 10 mg once daily, given with or without food.

Fish oil supplementation (Lovaza)

- Lovaza (omega-3-acid ethyl esters) is a prescription form of concentrated fish oil EPA 465 mg and docosahexaenoic acid 375 mg.
- The daily dose is 4 g/day (qd or bid)
- This product lowers triglycerides by 14% to 30% and raises HDL by about 10%.
- Side effect: thrombocytopenia and bleeding

Special Considerations

Elderly

- At a higher risk of developing CHD
- Risk decreases with lipid lowering therapy
- **Statins** the preferred therapy in older patients
- older patients are more likely to have
 - constipation (BARs)
 - skin and eye changes (niacin)
 - gout (niacin)
 - gallstones (fibric acid derivatives), and
 - bone/joint disorders (fibric acid derivatives, statins)

Special Considerations

Pregnancy

- Cholesterol and triglyceride levels rise progressively throughout pregnancy
- Triglyceride levels may go up by as much as 150 mg/dL (1.70 mmol/L)
- **Drug therapy is neither instituted nor usually continued during pregnancy**
- If the patient is **very high risk**: Consider BAR
- **Statins are category X and are contraindicated**
- Ezetimibe category C
- **Dietary therapy is the main stay of treatment**

Special Considerations

Children

- Drug therapy in children is not recommended until the age of 8 years or older
- **statins** are safe and effective in children and provide greater lipid lowering than BAR
- Severe forms of hypercholesterolemia (e.g., familial hypercholesterolemia) may require more aggressive treatment

Concurrent Disease States

- Nephrotic syndrome, end-stage renal disease, and hypertension
- **Statins** have been shown to be effective in reducing elevated total and LDL cholesterol in the nephrotic syndrome and renal insufficiency
- Need for combination use: Fibric acid derivatives + statins
- Bile acid resins do not correct the lipid abnormalities seen in renal insufficiency
- Diabetic dyslipidemia: Statins (Initial choice), fibrates and Niacine

Concurrent Disease States

Hypertension in patients with hypercholesterolemia

- Avoid the use of **thiazide diuretics** and **β -blockers**
- Bile acid sequestrants may bind to thiazide diuretics and some **β -blockers**, and may interfere with their absorption
 - **Give the antihypertensive 1 hour before or 4 hours after the resin**
- Niacin may magnify the hypotensive effects of vasodilators

Evaluation of Outcomes

- Monitor T-C, LDL-C, HDL-C, and TG for pts being treated
- Follow up interval is dependent on the severity of illness, and pts with known CAD or multiple risk factors should be monitored more closely
- Less commonly used laboratory measurements include C-reactive protein, homocysteine, apolipoprotein B, and lipoprotein T-C

Thank You for Attention